Data Science Case Study 3: Detecting Generator Usage from Sensor Data

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1. Introduction

1.1 Problem Statement

In many off-grid areas and regions with unreliable electricity supply, diesel generators are the primary source of power or provide essential backup capacity. In some African countries generator capacity has for a long time rivaled or exceeded grid supply (International Finance Corporation [IFC] & Schatz Energy Research Center, 2019). A study done between 2008 to 2011 on 13,140 healthcare facilities in 11 Sub-Saharan African countries actually showed that all hospitals relied on generators as their primary electricity source in 2008 and 2011 (Adair-Rohani et al., 2013). Fortunately, solar generation is increasingly emerging as a cleaner alternative, supported by international funding initiatives aimed at decarbonization and sustainable energy access (World Bank, 2020; Sustainable Energy for All [SEforALL], 2023). As a result, facilities draw electricity from varying mixes of grid, solar, and diesel generators. Accurately quantifying the impact of solarization—particularly reductions in carbon emissions and generator fuel expenditure—requires reliable data on generator versus solar usage durations.Yet, there is limited visibility into which source is active at a given time.

To address this, nLine has applied a **rule-based algorithm to minute-by-minute granular voltage and frequency data from sensors in healthcare facilities in Sierra Leone to detect the active power source**. This method is customized and fine-tuned for the Sierra Leone-specific case, making it difficult to generalize to other contexts where broader application would be useful. To overcome this, we propose a supervised machine learning (ML) approach that builds on the strengths of the rule-based method. While the existing algorithm achieves near-perfect results within Sierra Leone, its accuracy stems from overspecificity, limiting applicability elsewhere.

Our approach uses the outputs of the rule-based algorithm as labeled data to train an ML model. This strategy leverages the near–ground truth quality of the rule-based classifications while enabling the ML model to learn patterns in a generalizable and scalable manner. The trained model is then tested against the rule-based outputs to validate performance.

By doing so, we retain the accuracy of the existing method while producing a solution that is adaptable to new hospitals, regions, and countries. This improves the capacity to monitor generator reliance, assess solarization impacts, and estimate avoided fuel use and emissions across diverse contexts.

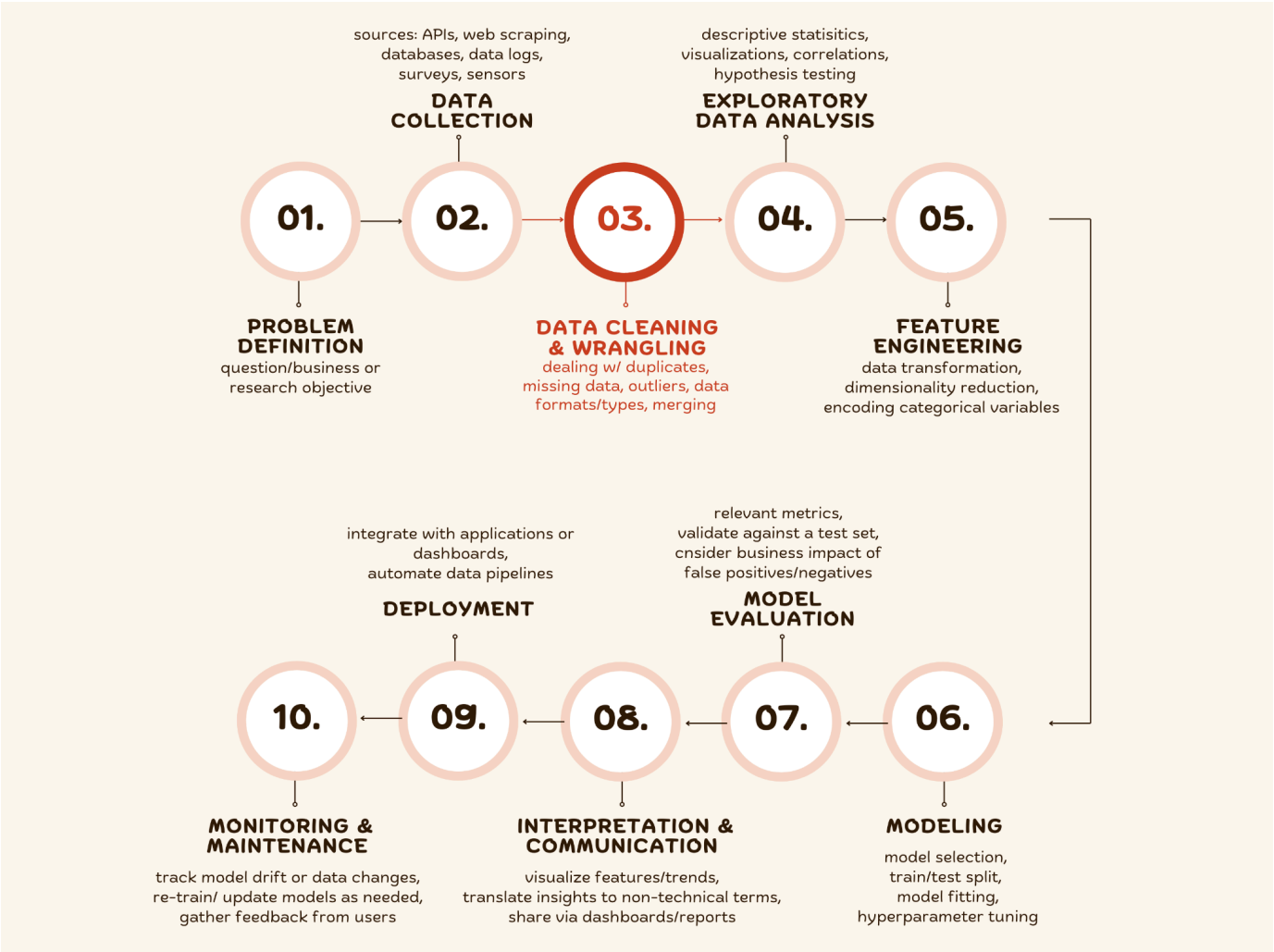
1.2 Data Source

The data used in this case study were collected by n-Line sensors deployed in different hospital rooms in healthcare facilities in Sierra Leone. nLine deployed GridWatch sensors across 17 government hospitals beginning in April 2023 as part of SEforALL's Sierra Leone Healthcare Electrification Project to enable SEforALL's former implementing project partner, Crown Agents, to quantitatively evaluate how well the newly installed renewable energy solutions address existing power quality and reliability challenges in these facilities. Installed in various rooms, including critical care areas, these sensors record high-frequency data on voltage, frequency, and outages every two minutes. This data provides a unique opportunity to detect active power sources with high temporal resolution, enabling quantification of generator reliance and solar utilization over time.

2. Review of the Rule-Based Power-Source Detection Algorithm

2.1 Overview of the Algorithm

The rule-based algorithm classifies each two-minute data point as solar, grid, generator, or outage based on frequency, derived features, and domain knowledge. It prioritizes expert insights over automated methods like unsupervised ML or Kalman filtering, which underfit due to data complexities. The algorithm uses thresholding on frequency and variability, augmented by ground-truth knowledge of room configurations and grid references.



2.2 Key Variables and Features Used

- Frequency: Nominal 50Hz for Sierra Leone's grid. Solar shows tight distribution around 50Hz; grid has moderate variance; generators vary widely (43-57Hz).
- Frequency Deviation: From the median frequency of the electrically closest grid-only room (threshold: ≤0.11Hz for grid).
- Frequency Spread/Variability: Mean Absolute Deviation (MAD) in a 15-minute sliding window (≤0.1 for solar).
- Room Power Source Label: Ground-truth configurations

2.3 Smoothing and Additional Considerations

Smoothing incorporates hospital-specific knowledge (e.g., from surveys or officials) and historical observations. Examples include restricting generator labels at PCMH to grid outages or setting Ola During to solar during grid-offs if within thresholds. Ongoing refinements add more smoothing as new insights emerge.

2.4 Strengths of the Rule-Based Algorithm.The approach developed for this study is characterized by its lightweight and interpretable design. It relies on simple rules and domain expertise, making it straightforward to implement, debug, and explain without requiring heavy computational resources. This ensures that the method remains accessible and transparent, which is particularly important when results must be communicated to non-technical stakeholders.

A key strength of the method lies in its robustness to data complexities. Unlike initial machine learning attempts that suffered from underfitting, this approach benefits from expert-tuned thresholds, allowing it to achieve higher reliability and accuracy. By embedding contextual knowledge, such as ground-truth room configurations and grid references, the system achieves a high level of precision in scenarios where reliable information is available. This not only enhances the accuracy of classification but also enables the calculation of detailed metrics, including fuel consumption.

The method is also designed to effectively handle challenging scenarios, particularly in cases where multiple sources are present within the same environment. While these multi-source rooms require more advanced processing, the approach simplifies the labeling of single-source rooms by automating the task. This balance between complexity management and automation increases overall efficiency.

Finally, the approach carries direct economic implications by supporting planning processes through the quantification of usage times. By accurately identifying energy source patterns, it provides actionable insights that can be used to inform resource allocation, improve operational planning, and ultimately contribute to cost savings and more sustainable energy use.

2.5 Shortcomings and Limitations

Despite its strengths, the approach also presents several limitations that constrain its broader applicability and long-term scalability. One major challenge is the lack of generalizability, as the method is tailored specifically to hospitals in Sierra Leone. Its reliance on assumptions such as nominal frequency values and characteristic grid behaviors makes it less adaptable to other regions without significant adjustments. For instance, variations in grid stability or generator usage patterns in different countries would require re-tuning of the system parameters.

Another limitation lies in its reliance on frequency as the dominant variable for classification. While frequency provides useful information, the underutilization of complementary sensor

data, such as voltage measurements, reduces the robustness of the model. Incorporating additional features could improve accuracy and resilience, especially in ambiguous or noisy conditions.

The method also employs informal smoothing techniques, which are based primarily on ad-hoc observations rather than rigorous statistical approaches. This reduces replicability and introduces a risk of bias, as the smoothing choices may not hold consistently across different contexts or datasets. Furthermore, the absence of mechanisms for handling uncertainty limits the model's reliability. Without confidence scores or probabilistic outputs, the system becomes brittle when confronted with overlapping frequency ranges, such as those shared between grid and generator sources.

Manual tuning of thresholds represents another constraint. For example, the 0.11 Hz deviation threshold was determined experimentally, which raises the possibility of overfitting to the current dataset. This reliance on human intervention not only reduces efficiency but also limits the model's adaptability to new or evolving conditions. Finally, scalability poses a significant concern. As deployments increase, maintaining hospital-specific rules becomes increasingly cumbersome. Without automation or adaptive learning mechanisms, the system may struggle to keep pace with larger-scale rollouts, reducing its practicality for widespread adoption.

While the rule-based method demonstrates clear advantages in interpretability, contextual accuracy, and immediate economic relevance, its limitations, particularly in scalability, generalizability, and robustness, highlight the need for more flexible solutions. These challenges naturally motivate the exploration of machine learning models, which can automate feature extraction, handle uncertainty, and adapt across diverse contexts with reduced reliance on manual tuning. The following section examines how machine learning approaches can address these gaps and extend the applicability of energy source detection systems.

3. The Machine Learning Approach

This section details the machine learning approach implemented to improve power source classification, moving beyond the limitations of the rule-based system. We engineered informative features from raw sensor data, trained multiple supervised models, and evaluated their performance to identify the most accurate and generalizable solution.

3.1 Available Features

The machine learning model is designed to integrate both raw sensor data and contextual metadata in order to achieve more accurate and adaptable source classification. The core numerical signals include instantaneous voltage and frequency measurements, which serve as primary indicators of power quality and stability. Voltage readings, expressed in volts, capture fluctuations that may distinguish between grid and generator sources, while frequency readings, measured in hertz, provide insights into nominal deviations that often characterize different supply modes.

In addition to these numerical inputs, the model incorporates contextual metadata to enhance interpretability and predictive performance. Specifically, the known power configuration for each room, such as “Solar with Grid Backup,” is treated as a categorical feature. This contextual information provides essential grounding for the model, allowing it to disambiguate cases where numerical signals alone may not be sufficient to distinguish between multiple possible sources. By combining quantitative sensor data with categorical context, the model establishes a more robust framework capable of handling variability across rooms and configurations.

3.2 Temporal Features (for EDA only):

Derived fields such as hour, day_of_week, and date were used exclusively for exploratory analysis and visualization and were not included in the final model feature set.

Target Variable:

- power_source_est: The rule-based estimate of the active power source, serving as the target label with classes: generator, grid, solar, and unknown.

Prior to feature engineering, the dataset was filtered to ensure robustness: Rows where power_source_est was labeled power_conditioner were removed.

A frequency band filter was applied, retaining only readings where frequency values fell between 43 Hz and 57 Hz, aligning with realistic operating conditions and removing extreme outliers.

3.2.1 Feature Engineering

To capture distinguishing characteristics of each power source, we engineered two key features from the raw frequency data:

Frequency Deviation from Grid Median:

Calculated as the absolute difference between a reading's frequency and the median frequency observed in grid-only rooms (approximately 50.1605 Hz). This feature helps identify readings that deviate from typical grid behavior.

Temporal Variability (30-minute Mean Absolute Deviation - MAD):

Computed using a sliding 30-minute window centered on each timestamp, this feature captures the short-term stability of the frequency signal. Solar sources, for instance, exhibit very low variability, whereas generators often show higher instability.

The final feature set used for model training included:

- ***Numerical features***: voltage, frequency, frequency_deviation_from_grid_median, MAD for 30 min window
- **Categorical feature**: power_source (one-hot encoded)

All numerical features were standardized using StandardScaler, and the categorical feature was encoded via OneHotEncoder within a ColumnTransformer to ensure consistent preprocessing.

3.3 Model Selection

We selected a diverse set of five supervised learning models to evaluate their effectiveness in capturing the underlying patterns in the data:

1. Logistic Regression: A linear baseline model.
2. Decision Tree: A non-linear model capable of capturing complex interactions.
3. Random Forest: An ensemble method to improve generalization and reduce overfitting.
4. Support Vector Machine (SVC): Effective for high-dimensional spaces with clear margins.
5. K-Nearest Neighbors (KNN): A non-parametric method suitable for capturing local patterns in the feature space.

This selection allowed us to compare both linear and non-linear approaches, with tree-based models and KNN expected to perform well given the engineered features and their ability to model non-linear decision boundaries.

3.4 Model Training

The target variable power_source_est was label-encoded into four classes. The dataset was split into training (80%) and testing (20%) sets. For the baseline models using only raw features (voltage and frequency), the split was stratified to maintain class distribution. For the enhanced feature set, a standard random split was applied.

All models were trained using default hyperparameters to establish a baseline performance level, with random states fixed for reproducibility where applicable.

3.5 Model Testing

Each model was evaluated on the held-out test set using weighted precision, recall, F1-score, and overall accuracy. Confusion matrices were also generated to analyze performance per class and identify specific misclassification patterns.

The evaluation was conducted in two phases:

- Baseline Performance: Using only raw voltage and frequency features.
- Enhanced Performance: Using the full engineered feature set with scaling and encoding.

3.6 Model Evaluation and Results

3.6.1 Baseline Models (Raw Features Only):

Models trained solely on voltage and frequency achieved moderate accuracy, with tree-based methods and KNN outperforming linear models:

- Decision Tree and Random Forest both achieved approximately 82.65% accuracy and F1-scores around 0.82.
- KNN followed closely with 80.84% accuracy.
- Logistic Regression lagged at 55.36%, highlighting the inadequacy of linear separability with raw features alone.

3.6.2 Enhanced Models (Engineered Features):

The inclusion of engineered features and proper preprocessing led to significant improvements across all models:

- Random Forest emerged as the top performer, achieving an accuracy of 89.98% and an F1-score of 80.91%.
- KNN was a close second, with an accuracy of 89.88% and an F1-score of 89.77%.
- Decision Tree, Logistic Regression, and SVC showed improved but lower performance compared to Random Forest and KNN.

Confusion matrices revealed that the enhanced models significantly reduced misclassifications, particularly between grid and solar categories, and better handled the unknown class.

Key Insights:

The engineered features provided critical signals that raw data alone could not capture.

The inclusion of the power_source configuration as a feature added valuable context, aligning with domain knowledge about room-level setups.

Tree-based ensembles and instance-based methods like KNN proved most effective in leveraging the non-linear relationships in the feature space.

This machine learning approach directly addresses several shortcomings of the rule-based algorithm:

- It moves beyond a reliance on frequency alone by incorporating voltage and stability metrics.

- It improves generalizability by learning patterns across all hospitals rather than relying on site-specific thresholds.
- The high accuracy and detailed confusion matrices provide a foundation for future integration of probabilistic outputs and hybrid decision-making frameworks.

Appendix

Data Exploration and Visualization

Data Cleaning and Wrangling

The data was first inspected to view the structure, columns, and data types contained in the data.

Cleaning steps:

- Converting time data to datetime for easy time-based analysis.
- Handled nulls: There were no null values in the data.
- Handling duplicates. The data had no duplicate rows.
- Standardize room/hospital names for consistency.

Data inspection and overview

The data was loaded and processed using Pandas. We have 7,982,001 rows and 12 columns of data, taken at 2-minute time stamps.

The dataset provided("sierra-leone_power_source_2023-2024.csv") includes time-series data with columns:

1. time

Timestamp of the reading (every 2 minutes). The time stamps from 1st May 2023 at 00:00:03 to 30th April 2024 at 23:58:15

2. *respondent_id*:

This is a unique identifier for the sensor or room being monitored. Readings were recorded from 66 sensors deployed in different rooms.

3. *room*:

Room description, including known power source configurations (e.g., "Main drug store - Solar w/Grid Backup"). This describes the room of the health facility where the sensor was deployed. The room is mapped to the power source supplying power. There are 57 unique rooms in the data set where the sensors were deployed.

4. *hospital*:

Hospital name. The data was recorded in 12 different hospitals in different regions across Sierra Leone. The hospitals include hospital: princess_christian, ola_during, masanga,

kambia, kabala, wilberforce, connaught, bonthe, hastings, king_harman_road, magburaka, and makeni.

5. *voltage*:

Voltage reading captured by the sensor.

6. *frequency*:

Frequency reading captured by the sensor.

7. *power_source*:

Known power source configuration per room. There are 8 unique power source configurations in the data source.

- Solar & Generator – Facility uses solar power with a generator as support/backup.
- Grid w/Solar & Generator Backup – Grid is primary, with both solar and generator as backups.
- Grid w/Generator Backup – Grid is primary, generator provides backup.
- Grid – Facility relies only on the national grid.
- Grid w/Solar Backup – Grid is primary, solar provides backup.
- Solar w/Grid Backup – Solar is primary, grid provides backup.
- Solar – Facility relies only on solar power.
- Generator – Facility relies only on a generator.

8. *power_source_est*:

An estimated active power source is detected by the current rule-based algorithm. Based on the current algorithm, the established power sources should result in solar, grid, or generator.

Duplicates and Null values

The dataset had no duplicate rows or columns. All the data entries were unique.The data also did not have any null values upon inspection

Standardizing room names for consistency.

The room names were mapped to the power source configurations. An example is “High Dependency Unit - Solar w/Grid Backup”. Since the power source configuration is duplicated in the power source column, we stripped the data for the room column to only record the specific room and the power source configuration in the power_source column.

Time-based analysis

All time-based analysis involved converting the time column to an appropriate datetime datatype for better time series visualisations and analysis. This also involved feature

engineering of time-based columns for better analysis. This time-based columns include, hour of the day, the day of the week, and the date of the day.

Exploratory Data Analysis (EDA)

In this section, we examine the cleaned dataset to uncover trends, patterns, and relationships that will guide subsequent modeling efforts. Specifically, our analysis focuses on:

- **Distribution and behavior of voltage and frequency:** Assessing how these variables vary within the dataset and identifying any notable patterns or anomalies.
- **Temporal variations in power sources:** Investigating how different power sources change over time, including variations by hour of the day and day of the week.
- **Feature–target relationships:** Exploring potential correlations and interactions between dataset features and the target variable, power_source_est.
- **Detection of unexpected findings:** Identifying anomalies, outliers, or unusual patterns that could influence feature engineering or model performance.

The insights from this step will help generate hypotheses, highlight potential data quality concerns, and inform the design of more effective machine learning models in the subsequent phase.

Visual Exploration of Frequency and Voltage

Relative power source configurations per hospital.

In this analysis, we look at the relative percentage usage of recorded power configurations in the hospital rooms.

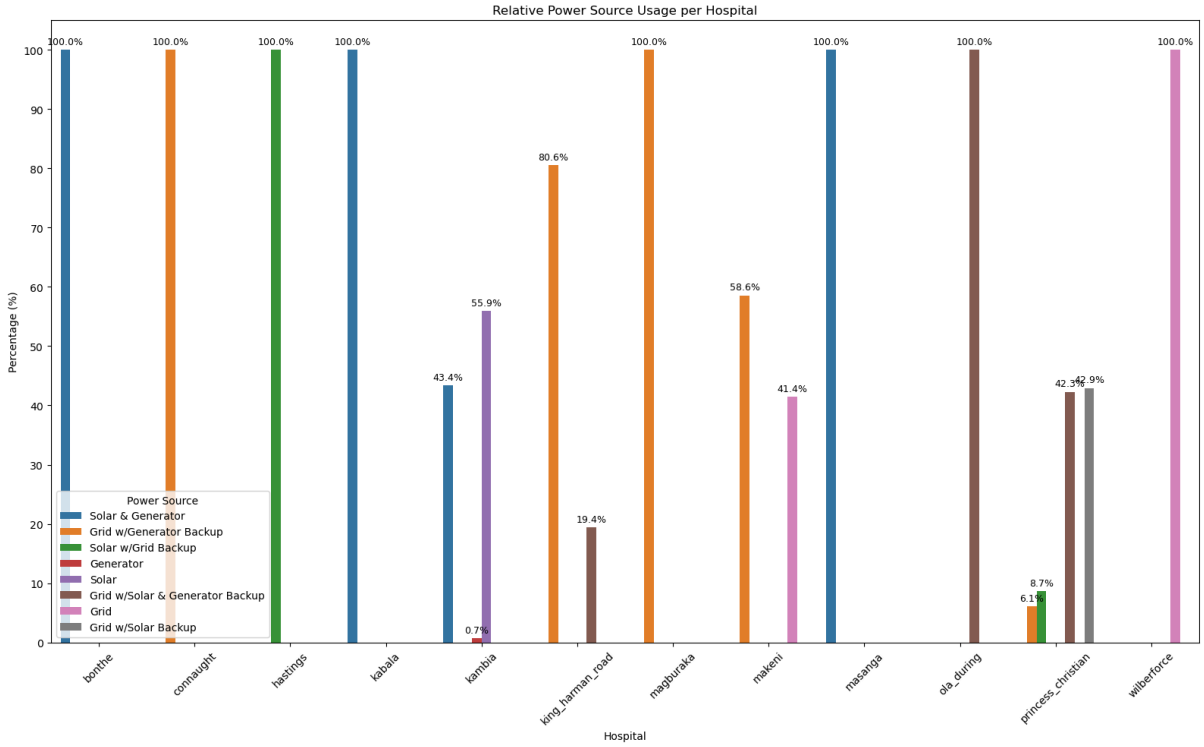


Fig 2: Power Usage Per hospital

Variety of Power Sources: The chart tracks multiple power sources, including Solar, Grid w/Generator Backup, Solar w/Grid Backup, Generator, Solar w/Generator Backup, Grid w/Solar & Generator Backup, and Grid w/Solar Backup, indicating diverse energy strategies across hospitals.

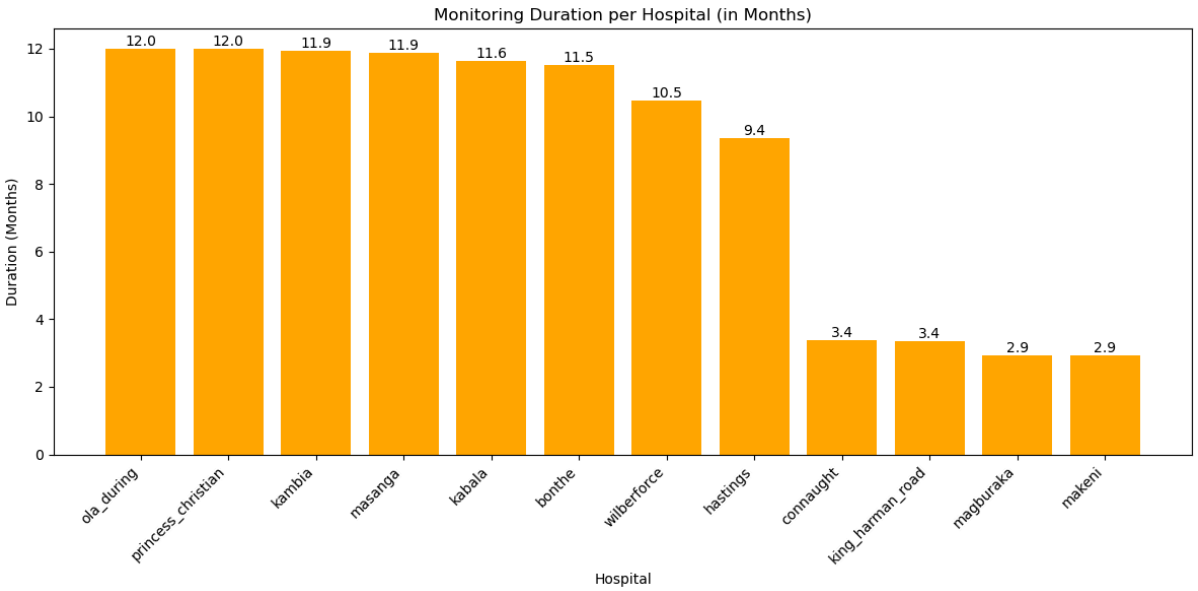
Dominant Power Sources: Some hospitals, like Bonthe, Connaught, Hastings, Kabala, and Masanga, show high usage (close to 100%) of a single power source (e.g., Solar or Grid w/Generator Backup), suggesting reliance on one primary energy solution.

Mixed Usage: Hospitals like Kambia, King Harman Road, Magburaka, Makeni, Ola During, Princess Christian, and Wilberforce exhibit varied usage of multiple power sources. For instance, Kambia uses Solar 55.9% and a small percentage of other sources, while Ola During and Princess Christian show significant reliance on Grid w/Solar Backup and Solar w/Generator Backup, respectively.

Low Diversity in Some Cases: Hospitals with near 100% usage of one source (e.g., Bonothe, Connaught) may lack redundancy, potentially making them vulnerable to power disruptions if that source fails.

Regional or Operational Differences: The variation in power source usage could reflect differences in location, infrastructure, funding, or energy availability, with some hospitals possibly in areas with reliable grid access and others relying on solar or generators.

Duration of Monitoring per hospital



- **Longest Monitoring:** *Ola_During, Princess_Christian, Kambia, and Masanga* recorded data for nearly a full year, making them the most consistently monitored hospitals.
- **Shortest Monitoring:** *Magburaka and Makeni* had the briefest monitoring spans, each lasting less than 3 months.
- **Intermediate Monitoring:** *Wilberforce and Hastings* were observed for about 9–10 months, placing them in the mid-range.
- **Overall Trend:** There is a noticeable drop-off in monitoring duration after the first seven hospitals, indicating uneven data coverage across sites.

Conclusion: These differences in monitoring duration can directly influence the interpretation of time-series analyses. Hospitals with shorter recording periods may appear to have sudden spikes or dominant trends that are actually a reflection of limited observation windows rather than true long-term behavior. In contrast, hospitals with longer monitoring periods provide a more stable and reliable picture of power source usage over time.

Distribution of power source configurations

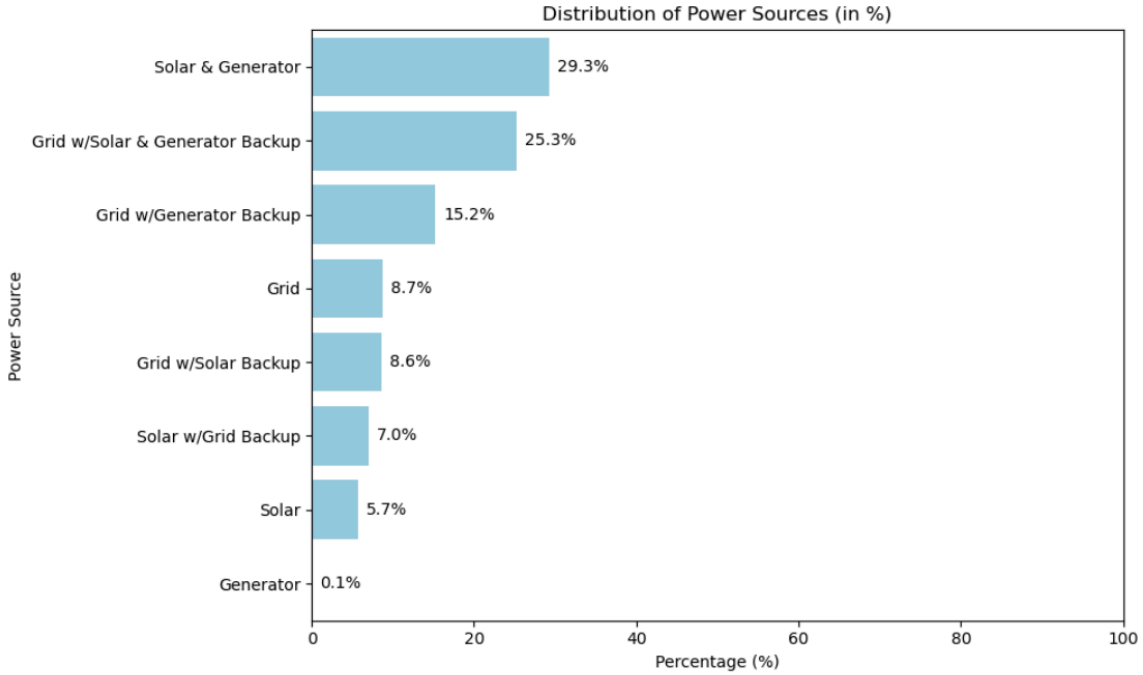


Fig 3: Distribution of power sources

1. **Most Common Sources**: The top power sources are Solar & Generator (29.3%) and Grid w/Solar & Generator Backup (25.3%), indicating a strong reliance on hybrid systems combining solar energy with backup options.
2. **Moderate Usage**: Grid w/Generator Backup (15.2%) and Grid (8.7%) are also significant, suggesting a notable dependence on grid-based systems with or without generator support.
3. **Lesser Usage**: Grid w/Solar Backup (8.6%), Solar w/Grid Backup (7.0%), and Solar alone (5.7%) are used to a lesser extent, showing limited standalone solar or grid-solar combinations.
4. **Minimal Usage**: Generator alone (0.1%) is almost negligible, indicating it's rarely used as a primary power source.

This distribution suggests a preference for hybrid power systems, likely for reliability, with solar and grid combinations dominating. The low use of standalone generators may reflect higher costs or maintenance issues.

Time-based Visualizations

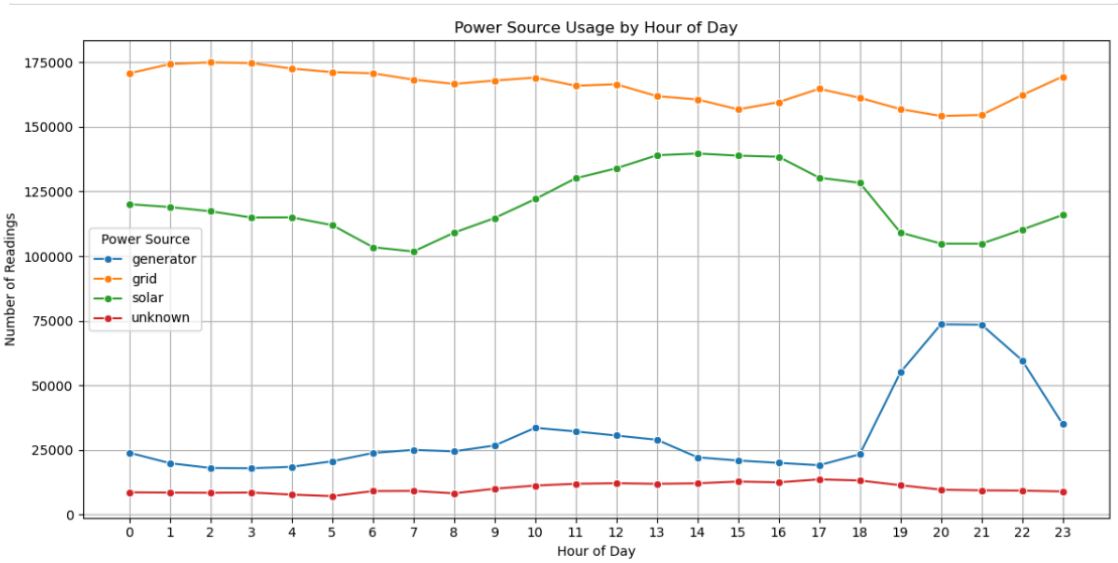


Fig 4: Power source usage by hour of day

1. **Grid Dominance**: The grid (orange line) consistently has the highest number of readings, ranging from about 15,000 to 17,500, with a peak around 1-3 AM and a slight dip around midday (12-2 PM), suggesting it's the primary power source throughout the day.
2. **Solar Usage**: Solar (green line) shows a moderate usage pattern, peaking around 13-15 PM at approximately 14,000 readings, likely due to daylight availability, and dropping to around 10,000 readings at night.
3. **Generator Activity**: Generator usage (blue line) is minimal during the day (around 25,000 readings), with a noticeable spike between 17-19 PM (up to 75,000 readings), possibly indicating backup activation during peak evening demand, before dropping off afterward.
4. **Unknown Readings**: The unknown category (red line) remains consistently low (around 10,000 readings) throughout the day, suggesting a small portion of data lacks clear power source classification.
5. **Daily Patterns**: The chart reflects a typical energy usage cycle, with the grid providing a stable baseline power, solar contributing during daylight hours, and

generators kicking in during high-demand evening hours, particularly around 6-7 PM (18-19 hours).

This indicates a system relying heavily on the grid, supplemented by solar during the day and generators during peak evening loads, with a small amount of unclassified usage.

Time series visualisation

Here, we look at the time series visualisation of solar power recorded data as it varies in the duration of recording.

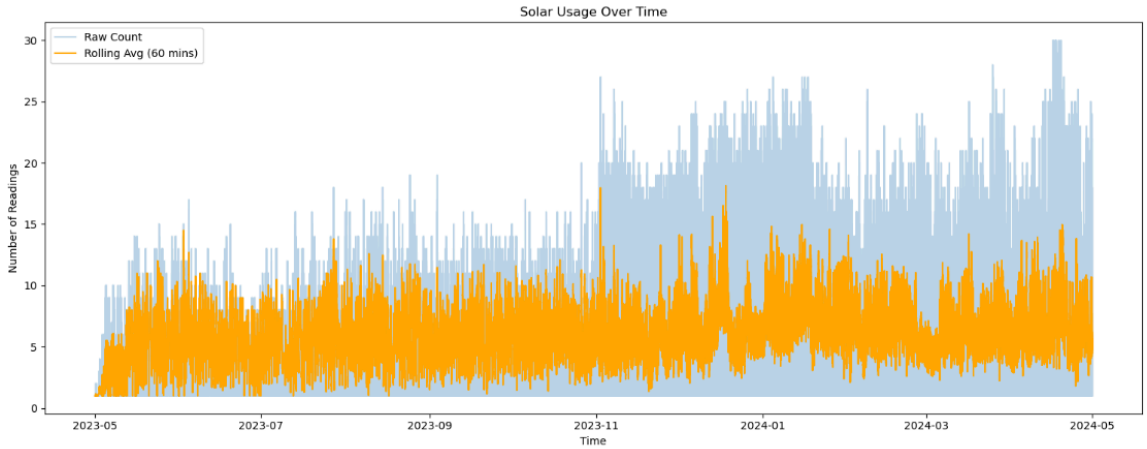


Fig 5: Solar Usage Over Time

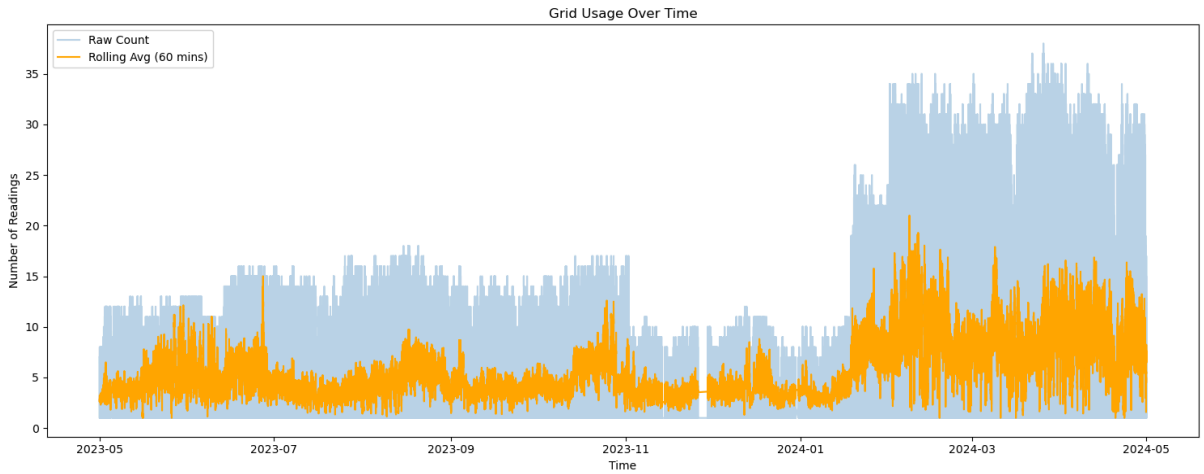


Fig 6: Grid Usage Over Time

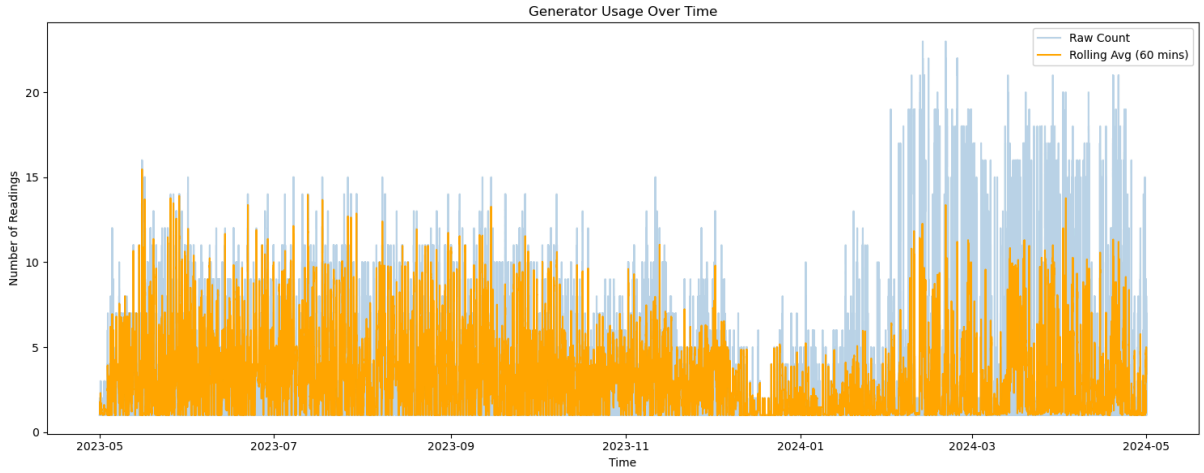


Fig 7: Generator Usage Over Time

Fig 6: Daily Frequency of Estimated Power Sources Over Time

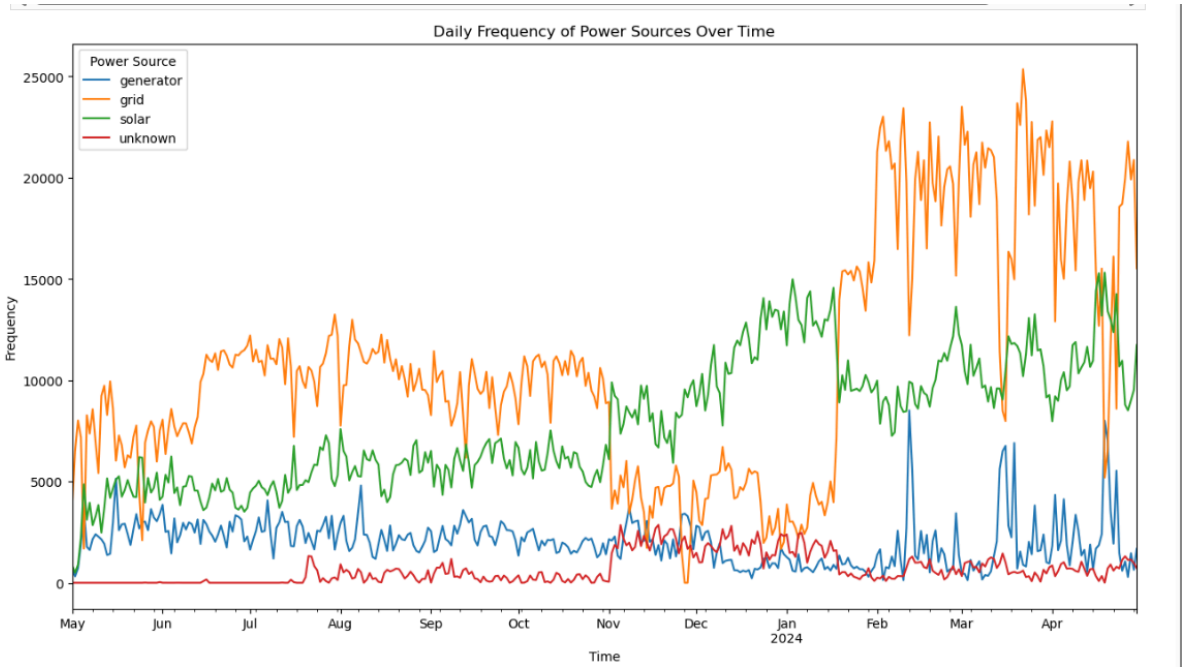


Fig 7: Daily Frequency of Power Sources Over Time

Time Series Analysis of Power Sources

The **generator usage** shows fluctuating activity across the period, with frequent short bursts of high usage. These spikes suggest irregular reliance on diesel generators, typically when other power sources were unavailable. The rolling average highlights sustained but intermittent use, with periods of both low and elevated activity.

The **solar usage** demonstrates a clearer, more consistent pattern, with moderate levels throughout 2023 and a gradual increase toward early 2024. The rolling average indicates stable usage trends, suggesting that solar provided a relatively steady share of the energy mix over time.

The **grid usage** displays the most pronounced shift. For most of 2023, grid reliance was relatively low and stable, but beginning in late January 2024, there was a sharp increase. Raw counts often exceeded 30 readings, with rolling averages surpassing 20, signaling a surge in grid activity and greater reliance on this source during early 2024.

Overall, the observed time series behavior is strongly influenced by the varying recording start dates across hospitals and their respective power source configurations. Hospitals that began recording later in the timeline contribute disproportionately to the sharp increases seen in early 2024. This highlights the importance of contextualizing usage patterns not just in terms of raw time series trends, but also in relation to the operational periods and configurations of the hospitals being monitored.

Voltage and Frequency Distribution.

Here we explore Frequency and Voltage visualisations.

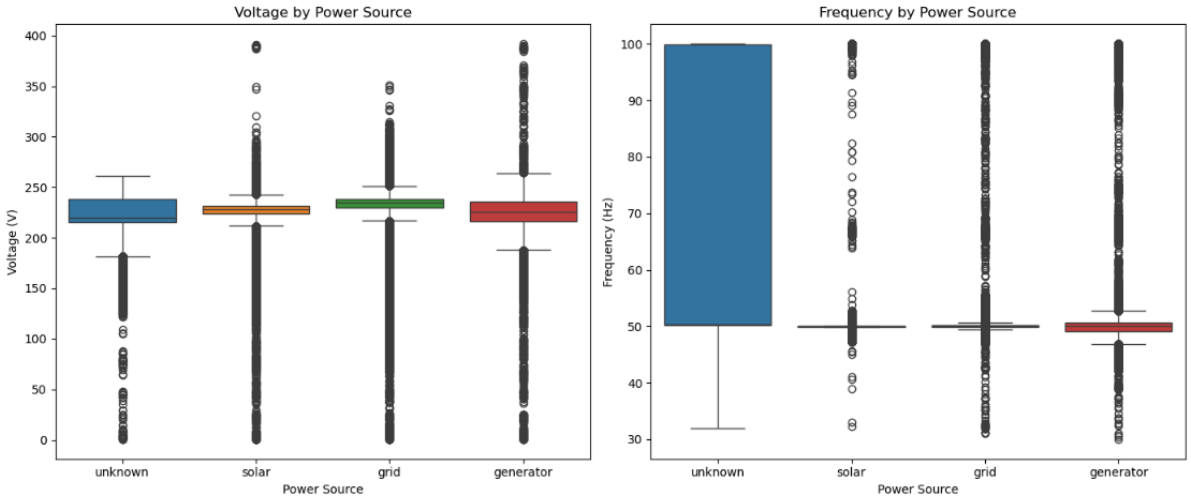


Fig 8: Voltage and Frequency by Estimated Power source

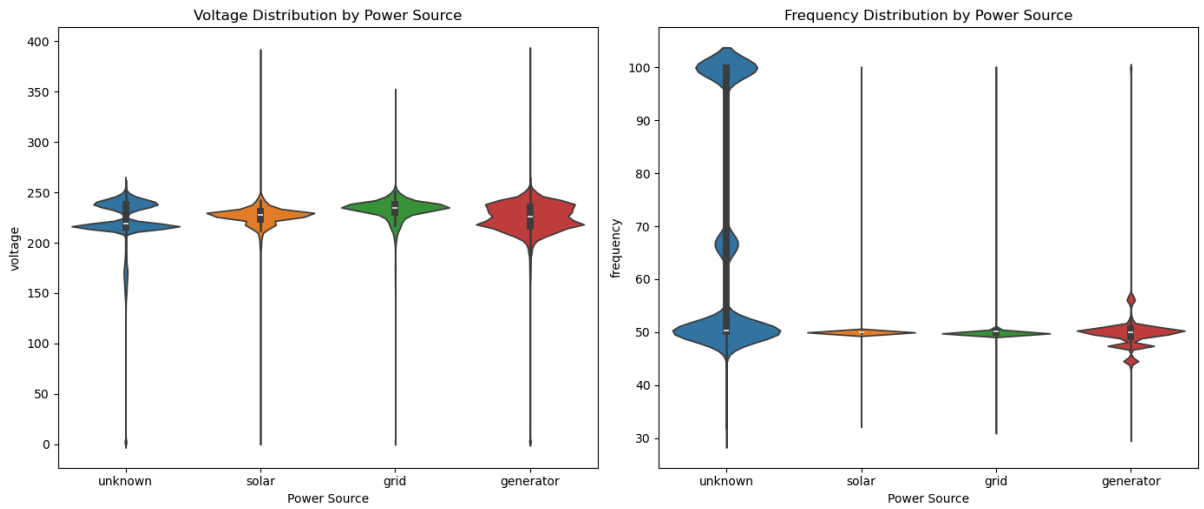


Fig 9:Violin plots on Voltage and Frequency by Estimated Power source

Voltage Distribution

The plot in Fig. 8 shows that the **grid** and **generator** power sources have a very stable voltage, with a tight distribution centered around **230 V**. The **solar** source also has a peak around 230 V but with a wider, flatter distribution, indicating more variability. The **unknown** power source shows a bimodal distribution, with one peak around 220 V and another around 240 V, suggesting it may represent a combination of different sources.

Frequency Distribution

The **grid**, **solar**, and **generator** sources all have a very stable frequency, with a sharp peak at **50 Hz**. In contrast, the **unknown** power source has a highly variable frequency, with distinct peaks at approximately **50 Hz**, **65 Hz**, and **100 Hz**, along with a wide spread of other values. This suggests that in the estimated power sources that result in 'unknown' most of the frequency values are outliers. The standard frequency range is ~45 Hz to 60 Hz. This is dependent on the power source and the region (continent where the frequency is being measured. The article "Frequency Control in a Power System" by Pietro Tumino (2020) provides a detailed look at how different levels of control maintain a stable frequency.

Correlations

Solar usage by hour and day of the week

This heatmap highlights the distinct daily and weekly patterns of solar power usage. Solar usage peaks consistently between 10 a.m. and 4 p.m., aligning with periods of maximum sunlight, while usage drops to zero overnight from around 8 p.m. to 6 a.m., as expected. Unlike grid power, solar usage shows remarkable consistency across the week, with similar daytime peaks from Monday through Sunday and no significant reduction on weekends. This uniformity underscores solar energy's dependence on natural daylight rather than variations in human activity or facility operations.

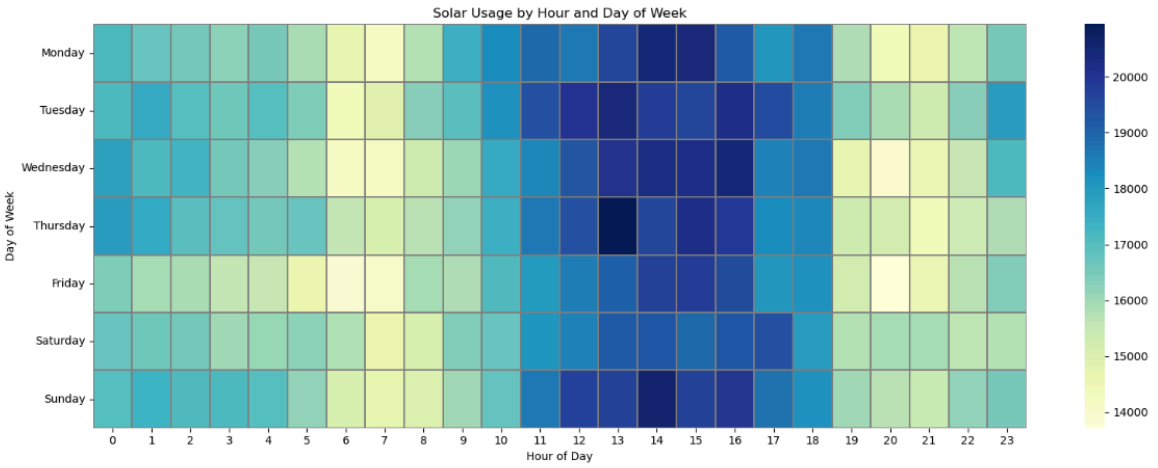
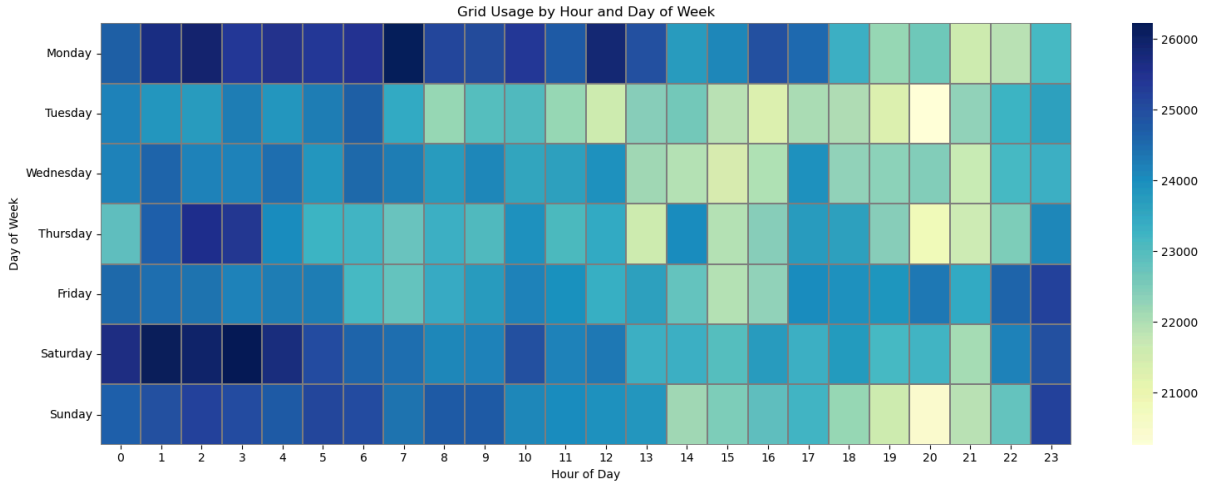


Fig 9: Solar Usage by hour or day of week

Grid usage by hour and day of the week

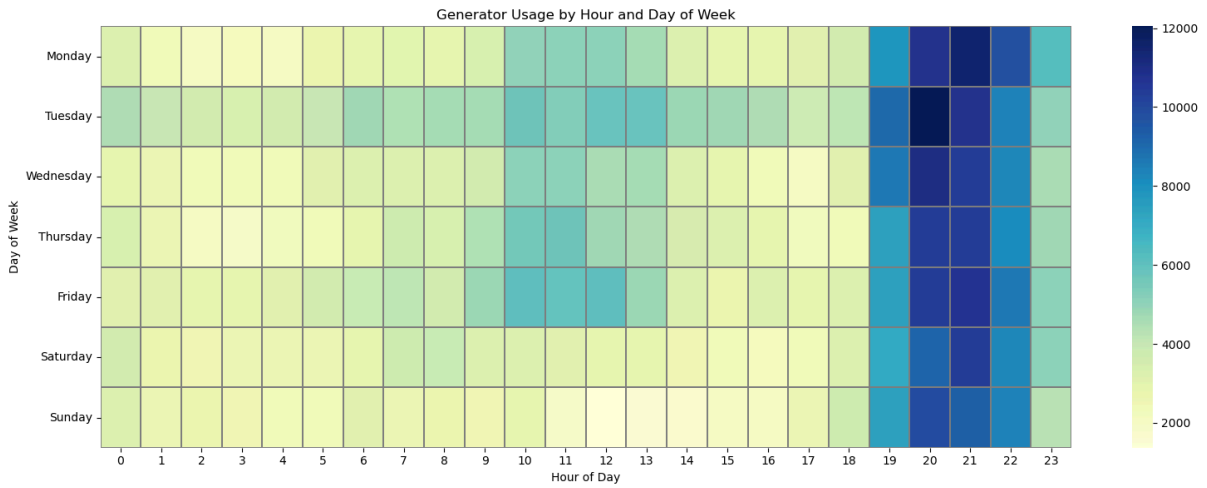
This heatmap illustrates patterns of grid power usage across hours of the day and days of the week, where lighter shades represent higher usage and darker shades indicate lower usage. Overall, grid usage is at its lowest during the early morning hours (midnight to 6 a.m.) and peaks during the daytime between 10 a.m. and 4 p.m., with a secondary rise in the evening from 6 p.m. to 9 p.m. Weekday trends (Monday to Friday) show consistently high usage during the day and evening, while weekends exhibit lower activity—Saturday being the lowest, particularly in the early morning and midday, and Sunday starting low but increasing in the afternoon and evening, resembling weekday patterns. The most prominent peaks occur on Tuesday and Wednesday in the late morning to early afternoon, and again on Friday evenings, highlighting clear temporal variations in grid reliance.



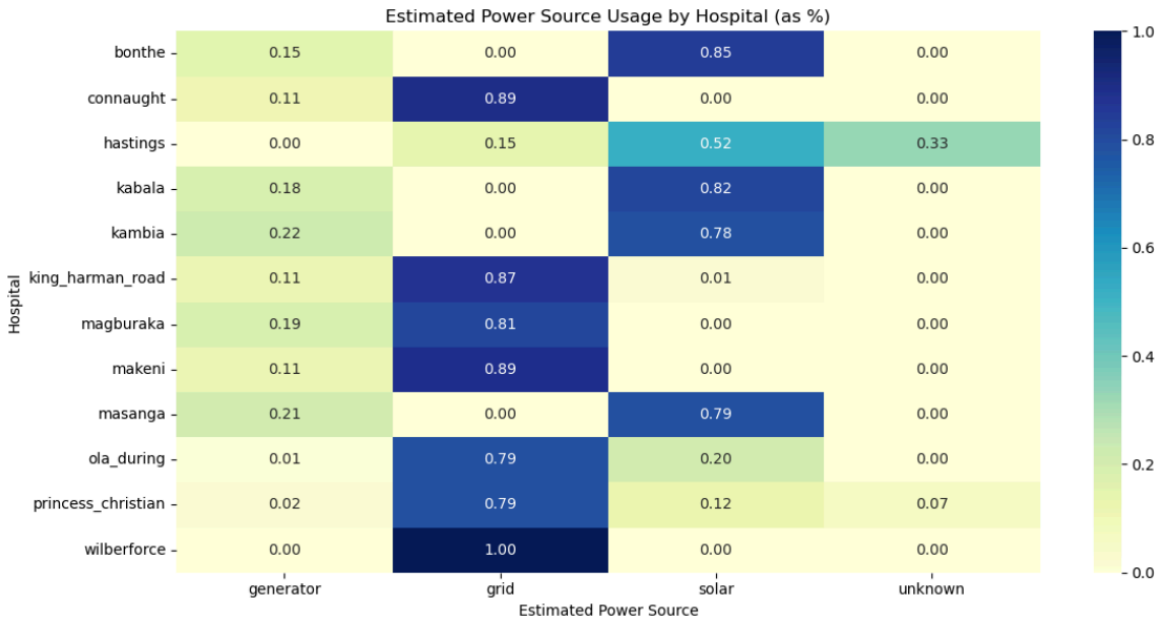
Generator usage by hour and day of the week.

Generator usage is highly concentrated in the evening hours, from roughly 6 p.m. to 10 p.m., regardless of the day of the week. This pattern is consistent across all seven days. The usage is highest between 8 p.m. and 10 p.m., as indicated by the darker blue cells.

There is minimal to no generator usage during the late night and early morning hours (1 a.m. to 7 a.m.). There is also a small, but noticeable, amount of generator activity during the daytime, specifically from about 9 a.m. to 4 p.m. This could represent times when the primary power source (likely the grid or solar) is unavailable or insufficient. The data suggests that generators are primarily used to supplement power during peak demand hours in the evening, rather than as a continuous power source throughout the day.



Correlation between the estimated power source and the hospital.



The heatmap of estimated power source usage across hospitals shows distinct patterns of energy dependence. Grid electricity is the dominant source in facilities such as Connaught, King Harman Road, Magburaka, Makeni, and Princess Christian, while Wilberforce relies entirely on the grid (100%). In contrast, Bonthe, Kabala, and Kambia are heavily dependent on solar, where it contributes more than 75% of their total supply. Some hospitals adopt mixed configurations: Hastings combines solar (52%), unknown (33%), and generator (15%), while Ola During blends grid (79%) and solar (20%). Generator usage overall is low, with Hastings showing the most notable reliance (15%) and others such as Bonthe, Kabala, Kambia, and Masanga contributing smaller shares.

Conclusion: These patterns highlight the uneven distribution of energy reliance across Sierra Leone's hospitals. While some facilities enjoy stable grid access, others depend heavily on solar or mixed systems to compensate for unreliable supply. Generator use remains supplementary, reinforcing the importance of investing in reliable solar solutions to improve resilience, reduce costs, and enhance healthcare delivery outcomes.

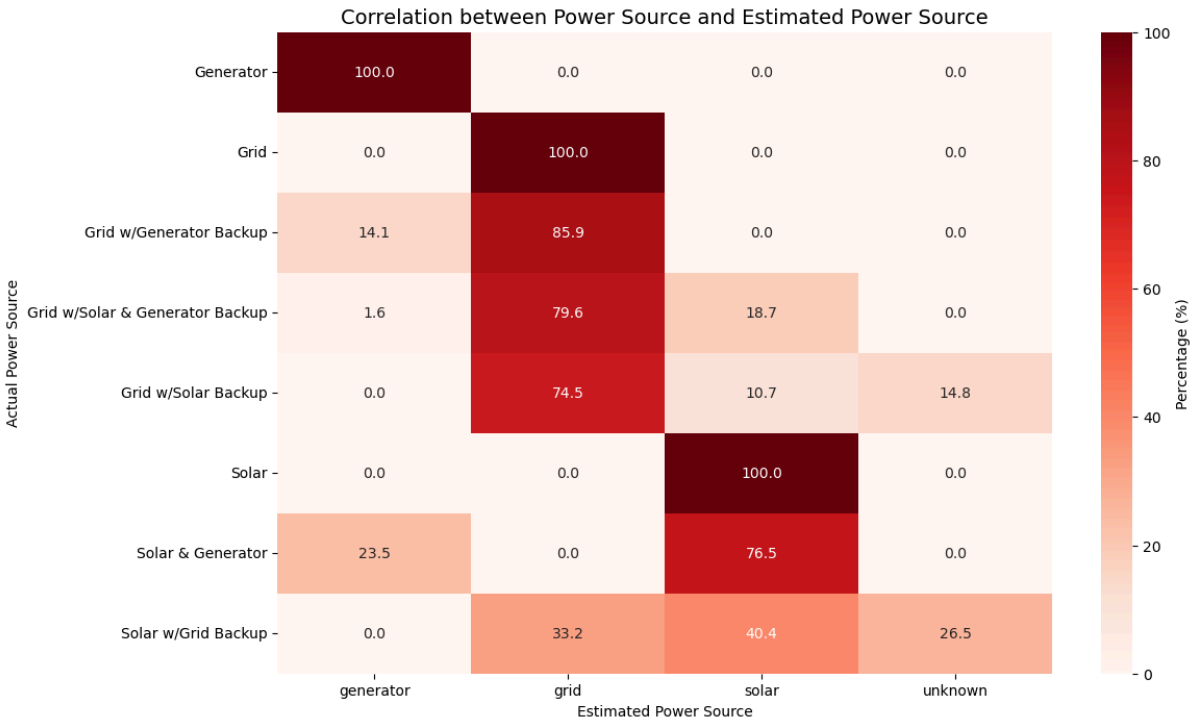


Fig 11: Relationship between power source and estimated power source

The heatmap (Fig. 11) illustrates the correlation between actual power source configurations and their estimated counterparts from the rule-based detection algorithm. Overall, the algorithm demonstrates *high accuracy for primary sources,* with perfect (100%) alignment when identifying Generator, Grid, and Solar individually. For *grid with backup systems*, performance is strong but not flawless—for example, a Grid with Generator Backup is correctly classified as Grid 85.9% of the time and as Generator 14.1%, reflecting occasional overlap in the data. Hybrid configurations pose greater challenges: Grid with Solar & Generator Backup is estimated as Grid (79.6%), Solar (18.7%), and Generator (1.6%), while Solar with Grid Backup shows a more scattered classification across Grid (33.2%), Solar (40.4%), and Unknown (26.5%). Similarly, Solar & Generator setups are estimated as Solar 76.5% of the time and Generator 23.5%. The **"Unknown"** category remains minimal overall, but its higher share in Solar with Grid Backup underscores the algorithm's uncertainty in complex hybrid systems.

Conclusion: While the rule-based approach performs reliably for single-source systems, its limitations become evident in mixed or backup configurations, where overlapping characteristics in the data make precise classification difficult. These insights highlight the need for more robust, data-driven methods to improve detection accuracy in hybrid energy environments.